Investigating Background Dependence in Deep Learning-Based Grape Leaf Disease Classification

Can a Grape Leaf Disease Classifier Recognize Disease Symptoms When Background Information Is Removed or Changed?

Research Area: Computer Vision, Deep Learning, Explainable AI, Model Robustness

Research Type: Experimental Research

Dataset: Niphad Grape Leaf Disease Dataset (NGLD)

Proposed Focus: Background Bias Detection, Foreground-Background Analysis, and Robust Disease Classification

Research Status: Preliminary experiments completed; proposed experiments and validation remain.

Abstract

Deep learning has demonstrated remarkable performance in image-based plant disease classification. However, high classification accuracy does not necessarily indicate that a model has learned the actual visual characteristics of plant diseases. A model may instead exploit unintended correlations between disease labels and image backgrounds, lighting conditions, leaf positioning, image composition, or other contextual information. Such behavior is commonly associated with shortcut learning and can negatively affect the reliability of deep learning models in real-world agricultural environments.

This research investigates whether deep learning-based grape leaf disease classifiers genuinely recognize disease-related visual characteristics or depend partially on background information. The study uses the Niphad Grape Leaf Disease Dataset (NGLD), which contains four categories: Healthy Leaves, Powdery Mildew, Downy Mildew, and Bacterial Leaf Spot.

Preliminary experiments were conducted using six deep learning architectures: ResNet-18, SqueezeNet 1.1, EfficientNet-B0, ViT-Tiny, DeiT-Tiny, and a proposed hybrid CNN-Transformer architecture named AGCTNet. The initial experimental pipeline incorporated image preprocessing, duplicate removal, near-duplicate detection, group-aware data splitting, class imbalance handling, and model evaluation.

A preliminary background-bias experiment was conducted using frozen ImageNet-pretrained ResNet-18 features and logistic regression. Three image conditions were evaluated: full images, centre-only images, and corners-only images. The corners-only experiment achieved 85.90% balanced accuracy, compared with 93.27% for full images. This indicates that substantial class-related information exists outside the central image region.

These findings raise concerns about contextual shortcuts in grape leaf disease classification. However, they do not independently prove that the trained deep learning classifiers rely on background information. Further controlled experiments are necessary.

The proposed research will investigate leaf-only classification, background-only classification, background replacement, background randomization, leaf segmentation, and cross-environment testing. It will also examine whether models trained with background diversification can maintain reliable classification performance when environmental conditions change.

The expected outcome is a systematic evaluation framework for understanding and reducing background dependence in grape leaf disease classification, with particular emphasis on improving model robustness and reliability in practical agricultural environments.

Keywords: Grape Leaf Disease Classification, Background Bias, Shortcut Learning, Computer Vision, Deep Learning, Image Segmentation, Model Robustness, Explainable AI, Domain Generalization.

1. Introduction

1.1 Background

Agricultural productivity is strongly influenced by plant health. Plant diseases can reduce crop quality, decrease production, and cause considerable economic losses. Early identification of diseases is therefore important for effective agricultural management.

Traditional plant disease identification often depends on visual inspection by farmers, agricultural specialists, or plant pathologists. Although this approach can be effective, it may require considerable experience and time, particularly when multiple diseases exhibit similar visual symptoms.

Recent developments in Artificial Intelligence (AI), particularly Deep Learning (DL) and Computer Vision (CV), have introduced automated approaches for plant disease classification.

Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and hybrid deep learning architectures can learn visual representations from images and classify different plant diseases.

For example, a grape leaf disease classification system may distinguish between:

- Healthy grape leaves.
- Powdery Mildew.
- Downy Mildew.
- Bacterial Leaf Spot.

A conventional classification system receives a leaf image and predicts its disease category.

However, an important question remains:

Does the model actually recognize the disease symptoms, or does it learn other visual patterns that happen to be associated with the disease labels?

This question is particularly important when deep learning models are intended for real-world agricultural applications.

1.2 Understanding the Background Problem

Consider two photographs of the same diseased grape leaf.

In the first photograph, the leaf is placed against a white background.

In the second photograph, the same leaf is photographed against soil, grass, or another plant.

The disease symptoms remain unchanged. Only the surrounding environment changes.

A reliable disease classifier should ideally predict the same disease in both photographs.

However, a model that has learned correlations between disease categories and background characteristics may produce different predictions.

For example:

- Powdery Mildew images may frequently contain a particular background.
- Downy Mildew images may have been photographed under different lighting conditions.
- Healthy leaves may have a more consistent image composition.
- A particular disease category may contain similar surroundings across many images.

A deep learning model can learn these correlations because they help minimize classification errors during training.

The model may consequently achieve high accuracy without learning sufficiently reliable disease-specific features.

This is an example of **shortcut learning**.

Shortcut learning occurs when a machine learning model uses easily available but potentially unreliable features instead of learning the intended underlying relationship.

In this research, the intended relationship is between the visual symptoms of grape leaf diseases and their corresponding disease categories.

1.3 Why This Problem Matters

A model that depends on background information may perform well on images similar to its training dataset but fail when deployed in different environments.

Consider the following situations:

1. A farmer photographs a diseased grape leaf in a vineyard.
2. Another farmer photographs the same disease against a different background.
3. The image is captured under different lighting conditions.
4. The leaf is photographed from a different angle.
5. The surrounding leaves, soil, branches, or objects become visible.

If the model depends heavily on contextual information, its predictions may become unreliable.

Therefore, high classification accuracy alone is insufficient to establish that a disease classification model has learned meaningful disease characteristics.

This research investigates whether the model can maintain its predictions when background information is removed or changed.

2. Problem Statement

Existing deep learning-based plant disease classification systems are commonly evaluated using classification accuracy, precision, recall, and F1-score.

Although these metrics measure predictive performance, they do not directly establish which image regions contribute to the model's predictions.

A model may classify grape leaf diseases by using:

- Disease lesions.
- Leaf color and texture.
- Leaf shape and structure.
- Background colors.
- Lighting conditions.
- Image composition.
- Other environmental characteristics.

Consequently, a model may achieve high accuracy while depending partially on irrelevant contextual information.

In the existing grape leaf disease classification project, preliminary experiments produced an unexpectedly high classification performance using image regions outside the central portion of the leaf image.

This suggests that additional investigation is necessary to determine whether background information contributes to disease classification.

The main research problem is therefore:

To investigate whether deep learning-based grape leaf disease classifiers rely primarily on disease-related visual characteristics or exploit background and contextual information, and to evaluate whether their classification performance remains reliable when background information is removed or modified.

3. Research Aim and Objectives

3.1 Main Aim

The primary aim of this research is to investigate and reduce background dependence in deep learning-based grape leaf disease classification by evaluating model performance under different foreground and background conditions.

3.2 Specific Objectives

The research will pursue the following objectives:

1. Investigate the presence of background-related information in grape leaf disease images.
2. Determine whether background regions contain information associated with different disease categories.
3. Evaluate existing deep learning models using original, leaf-only, and background-only images.
4. Investigate whether removing background information affects disease classification performance.
5. Examine the effect of replacing original backgrounds with different backgrounds.
6. Evaluate whether a model maintains consistent predictions when the same leaf is presented in different environmental conditions.
7. Investigate whether background randomization during training can reduce dependence on contextual information.
8. Compare CNN, Transformer, and hybrid architectures under different background conditions.
9. Use explainability techniques to investigate whether models focus on actual leaf regions and disease symptoms.
10. Evaluate the generalization ability of trained models using independently collected grape leaf images.

4. Research Questions

The research is guided by the following questions.

RQ1: Can a grape leaf disease classifier accurately identify diseases when the background is completely removed?

RQ2: Can background regions alone provide sufficient information for predicting grape leaf disease categories?

RQ3: How does replacing the original background with an unrelated background affect classification performance?

RQ4: Do different deep learning architectures exhibit different levels of background dependence?

RQ5: Can background randomization during training improve classification robustness?

RQ6: Does a model trained with background diversification focus more consistently on disease-related leaf regions?

RQ7: Can the improved model maintain its performance on independently collected grape leaf images?

5. Research Hypotheses

The following hypotheses will be investigated experimentally.

H1: Background Information Hypothesis

Background regions contain class-associated information that can contribute to grape leaf disease classification.

H2: Leaf-Only Classification Hypothesis

A model trained and evaluated using leaf-only images can learn meaningful disease-related characteristics without requiring the original background.

H3: Background Replacement Hypothesis

Changing the background while preserving the original leaf will affect the predictions of models that have learned background-related shortcuts.

H4: Background Randomization Hypothesis

Training with randomized backgrounds will reduce the model's dependence on environmental information.

H5: Architecture Sensitivity Hypothesis

Different deep learning architectures will demonstrate different levels of sensitivity to background modifications.

H6: Robustness Hypothesis

A model with reduced background dependence will maintain more consistent predictions across different environmental conditions.

These hypotheses are proposed research questions, not established findings.

6. Literature Background

6.1 Shortcut Learning in Deep Learning

Shortcut learning describes situations in which deep learning models exploit predictive patterns that are not necessarily related to the intended task.

Geirhos et al. (2020) discussed how neural networks can achieve strong benchmark performance using shortcuts that fail under more challenging testing conditions.

For example, a classifier may associate a particular background with a disease category rather than learning the actual visual characteristics of the disease.

This research applies the shortcut learning concept to grape leaf disease classification.

6.2 Background Information in Plant Disease Classification

Plant disease classification research has previously investigated the importance of background information.

Mohanty et al. (2016) evaluated plant disease classification using original, grayscale, and segmented versions of the PlantVillage dataset.

Their experiments demonstrated the importance of examining the effect of image background information when evaluating plant disease classifiers.

Other research has also investigated the difficulty of transferring plant disease recognition models from controlled environments to natural field conditions.

For example, Beery et al. (2018) demonstrated that strong performance within familiar environments does not guarantee successful generalization to unfamiliar environments.

6.3 Domain Generalization in Plant Disease Recognition

Domain generalization investigates whether a trained model can perform reliably when the testing data comes from a different distribution than the training data.

In agricultural applications, distribution changes can occur because of:

- Different cameras.
- Different farms.
- Different backgrounds.
- Seasonal changes.
- Lighting variations.
- Different leaf orientations.
- Different image acquisition conditions.

Recent work on domain-generalized plant leaf disease recognition has specifically investigated the effects of background and visual style differences.

This provides a strong research foundation for investigating background dependence in grape leaf classification.

6.4 Research Gap

The existing literature establishes that background information and environmental changes can affect plant disease classification.

Therefore, simply demonstrating that background-only images achieve above-chance classification accuracy would not be sufficient to establish a new research contribution.

The proposed study will focus on a more specific experimental investigation:

1. A grape-specific background-dependence analysis using NGLD.
2. Controlled experiments involving the removal and replacement of backgrounds.
3. Paired prediction consistency tests using the same leaf under different backgrounds.
4. Comparison of multiple CNN, Transformer, and hybrid architectures.
5. Background-randomized training and evaluation.
6. Investigation of whether models recognize disease symptoms rather than merely leaf-level or environmental characteristics.
7. Independent validation using newly collected grape leaf photographs.

The contribution will depend on the quality of these experiments and the results obtained.

7. Dataset Description

7.1 Dataset Selection

This research uses the Niphad Grape Leaf Disease Dataset (NGLD).

The dataset contains 2,726 images of grape leaves divided into four disease categories.

The images were collected using mobile phones in grape-growing environments in the Niphad region of Maharashtra, India.

The dataset is publicly available through Mendeley Data.

Dataset DOI:

10.17632/8nnd2ypcv3.5

7.2 Original Dataset Distribution

Class	Number of Images
Healthy Leaves	1,254
Powdery Mildew	406
Downy Mildew	966
Bacterial Leaf Spot	100
Total	**2,726**

The notebook currently uses the class name Bacterial Rot, although the original dataset documentation identifies the 100-image category as Bacterial Leaf Spot.

This terminology should be verified with the dataset documentation and a plant pathology expert before final publication.

Similarly, the notebook uses the spelling Downey Mildew. The correct disease name is Downy Mildew.

7.3 Dataset Characteristics

The dataset presents several challenges:

- Significant class imbalance.
- Different visual characteristics across disease categories.
- Potentially correlated background information.
- Similar or duplicate images.

- Limited examples of Bacterial Leaf Spot.
- Possible differences in image acquisition conditions.

These characteristics make it appropriate for investigating model robustness and background dependence.

7.4 Dataset Cleaning Already Completed

The existing notebook implemented:

- Exact duplicate detection using pixel-level MD5 fingerprints.
- Near-duplicate detection using perceptual hashing.
- ResNet-18 embedding similarity.
- Group construction using connected components.
- Removal of cross-class similarity groups.

The cleaning results were:

Cleaning operation	Result
Original images	2,726
Exact duplicate copies removed	321
Images in cross-class similarity groups removed	75
Images retained	2,330

The final dataset was divided into training, validation, and testing groups.

Split	Number of Images
Training	1,878
Validation	219
Testing	233
Total	**2,330**

The current notebook reports no shared near-duplicate groups across the three splits.

This is important because images of the same or highly similar leaf should not appear in both training and testing data.

However, the duplicate thresholds and similarity clusters should also be manually audited to ensure that unrelated leaves have not been grouped together.

8. Preliminary Experimental Findings

Before conducting the proposed background-removal experiments, the existing notebook produced several useful findings.

These findings motivate the proposed research but do not yet answer the main research question conclusively.

8.1 Baseline Model Development

Six deep learning architectures were evaluated.

Model	Architecture Type
ResNet-18	CNN
SqueezeNet 1.1	CNN
EfficientNet-B0	CNN
ViT-Tiny	Vision Transformer
DeiT-Tiny	Vision Transformer
AGCTNet	Hybrid CNN-Transformer

AGCTNet combines convolutional and transformer-based feature extraction.

The notebook uses the following techniques:

- CLAHE-based contrast enhancement.
- Gaussian smoothing.
- Image augmentation.
- Focal loss.
- Weighted sampling.
- Transfer learning.
- Learning-rate selection.
- Early stopping.

8.2 Baseline Test Results

The following results were obtained from the current notebook.

Model	Test Accuracy	Macro-F1
ResNet-18	98.28%	97.45%
SqueezeNet 1.1	98.28%	97.24%
EfficientNet-B0	98.71%	97.72%
ViT-Tiny	97.85%	95.81%
DeiT-Tiny	98.28%	97.46%
AGCTNet	98.71%	97.72%

AGCTNet and EfficientNet-B0 achieved the highest reported test accuracy.

However, they achieved identical accuracy and macro-F1 values.

Therefore, the current results do not demonstrate that AGCTNet is superior to EfficientNet-B0.

8.3 Cross-Validation Results

The notebook also conducted five-fold group-aware cross-validation within the development dataset.

Model	Mean Accuracy	Mean Macro-F1
ResNet-18	98.47%	97.28%
EfficientNet-B0	98.23%	97.15%
AGCTNet	97.47%	96.45%
DeiT-Tiny	97.47%	96.27%
ViT-Tiny	96.89%	95.09%
SqueezeNet 1.1	95.28%	92.84%

ResNet-18 achieved the highest mean macro-F1.

This indicates that high performance on a single holdout test set does not necessarily imply superior performance across different data partitions.

The cross-validation procedure still needs a fully nested model-selection protocol because the selected learning rates were determined using the original development validation set before the folds were evaluated.

Consequently, these cross-validation results should be described as preliminary rather than fully independent nested-CV estimates.

9. Preliminary Background-Bias Experiment

9.1 Purpose

The existing notebook contains a preliminary experiment designed to investigate whether image regions outside the centre contain information associated with grape leaf disease categories.

The experiment used:

- ImageNet-pretrained ResNet-18 as a frozen feature extractor.
- Logistic Regression as the classification model.
- Training images for model fitting.
- Testing images for evaluation.

Three image conditions were created.

Condition A: Full Image

The original preprocessed image was used without masking.

Condition B: Centre-Only Image

Only the central region of the image was preserved.

The remaining borders were replaced with a constant grey value.

Condition C: Corners-Only Image

The four corner regions were preserved.

The remaining image area was replaced with a constant grey value.

The purpose was to investigate whether disease-associated information exists in the image background and surrounding regions.

9.2 Experimental Results

The notebook produced the following results:

Image Condition	Accuracy	Balanced Accuracy	Macro-F1
Full Image	94.42%	93.27%	94.41%
Centre-Only	93.56%	92.36%	93.13%
Corners-Only	90.99%	85.90%	88.90%

For a four-class balanced classification problem, the expected balanced accuracy of uniform random guessing is 25%.

The corners-only experiment achieved 85.90% balanced accuracy.

This is a substantial result and deserves further investigation.

9.3 Interpretation

The corners-only experiment demonstrates that the image regions outside the central area contain substantial information associated with the disease labels.

This information may include:

- Background colors.
- Environmental characteristics.
- Image composition.
- Leaf positioning.
- Remaining leaf fragments.
- Lighting and acquisition patterns.

The result raises the possibility that the dataset contains contextual shortcuts.

However, an important distinction must be maintained.

The experiment uses a separate frozen ResNet-18 feature extractor and logistic regression classifier. It does not directly measure how much the six trained disease classifiers rely on background information.

Furthermore, the corners-only images may still contain parts of the leaf.

Therefore, the experiment is evidence of contextual predictability, not conclusive proof of background dependence in the original classifiers.

9.4 Main Research Motivation

The preliminary findings establish the motivation for the proposed investigation:

If image regions outside the central leaf region can predict disease categories, can the actual disease classifier maintain its performance when those contextual regions are removed or replaced?

This is the central question that the next experimental phase will investigate.

10. Proposed Research Methodology

The proposed methodology consists of seven major stages.

Stage 1: Image Segmentation and Leaf Isolation

Objective

Separate the grape leaf from the surrounding background.

Why is this necessary?

The current dataset contains complete images.

To understand whether the model recognizes disease characteristics, it is necessary to distinguish the leaf region from the surrounding environment.

A binary segmentation mask will be generated for each image.

The mask will identify:

- Leaf pixels.
- Background pixels.

A binary mask can be represented as:

$M(x,y) = 1$ if the pixel belongs to the leaf

$M(x,y) = 0$ if the pixel belongs to the background

The original image is represented by $I(x,y)$.

The leaf-only image can be constructed as:

$I_{leaf}(x,y) = I(x,y) \times M(x,y)$

The background-only image can be constructed as:

$$I_{background}(x,y) = I(x,y) \times (1 - M(x,y))$$

The background should be replaced with a controlled neutral color rather than simply assuming that black pixels are equivalent to the original background.

Suggested Segmentation Techniques

The following techniques can be investigated:

1. OpenCV GrabCut.
2. Color-based segmentation.
3. U-Net segmentation.
4. Segment Anything Model (SAM).

A manually annotated subset should be used to evaluate segmentation quality.

Automated segmentation must be checked carefully because diseased regions may have unusual colors or textures that segmentation algorithms could incorrectly remove.

Expected Output

For each original image, generate:

- Original image.
- Binary leaf mask.
- Leaf-only image.
- Background-only image.

Stage 2: Construction of Experimental Image Conditions

After segmentation, construct multiple versions of the same image.

The same leaf should be used across the different conditions so that changes in prediction can be attributed to image manipulation rather than a change in disease sample.

Condition A: Original Images

Use the original images with their existing backgrounds.

Purpose: Establish baseline performance.

Condition B: Leaf-Only Images

Preserve the complete leaf and replace the background with a neutral color.

Purpose: Investigate classification using leaf information without the original background.

Condition C: Background-Only Images

Remove the leaf and preserve only the background.

Purpose: Investigate whether background information alone predicts disease categories.

Condition D: Background-Replaced Images

Preserve the leaf but replace the original background with another background.

Possible backgrounds:

- White.
- Black.
- Light grey.
- Green.
- Soil.
- Grass.
- Other natural environmental scenes.

Purpose: Investigate the sensitivity of predictions to background changes.

Condition E: Randomized Background Images

Replace the original background with randomly selected backgrounds.

The background selection should be independent of the disease label.

Purpose: Reduce correlations between disease categories and specific backgrounds.

Condition F: Background-Modified Images

Apply controlled changes to the original background:

- Gaussian blur.
- Brightness changes.
- Contrast changes.
- Color shifts.
- Texture changes.

- Background noise.

Purpose: Evaluate robustness to environmental changes.

Experimental Image Summary

Condition	Leaf	Original Background	Modified Background
Original	Present	Present	No
Leaf-only	Present	Removed	Neutral
Background-only	Removed	Present	No
Background-replaced	Present	Removed	New background
Randomized background	Present	Removed	Random
Background-modified	Present	Present	Modified

All transformations must preserve the disease-related visual information as much as possible.

Stage 3: Baseline Model Evaluation

The six existing models will be evaluated using the original images.

Their baseline performance will be recorded before testing background interventions.

The models are:

- ResNet-18.
- SqueezeNet 1.1.
- EfficientNet-B0.
- ViT-Tiny.
- DeiT-Tiny.
- AGCTNet.

The same training, validation, and testing groups should be maintained.

Model selection must be performed using training and validation data only.

The final test set must remain untouched until the experimental configurations have been finalized.

Stage 4: Background Dependence Evaluation

This stage is the central part of the research.

Each trained model will receive multiple versions of the same test image.

For example:

Original image → Model → Prediction A

Leaf-only image → Model → Prediction B

Background-replaced image → Model → Prediction C

Background-only image → Model → Prediction D

The predictions will be compared.

Example

Suppose the original image contains a grape leaf affected by Powdery Mildew.

Image Condition	Model Prediction
Original image	Powdery Mildew
Leaf-only image	Powdery Mildew
White background	Powdery Mildew
Soil background	Powdery Mildew
Grass background	Downy Mildew
Background-only image	Powdery Mildew

This hypothetical example would suggest that the model's prediction changes under a background intervention.

The background-only prediction would also indicate that contextual information may be associated with the disease label.

The experiment must be repeated across the complete test set before drawing conclusions.

Stage 5: Background Randomization During Training

After measuring the existing models' background dependence, investigate whether training with randomized backgrounds improves robustness.

Two main training strategies will be compared.

Strategy A: Standard Training

Train the model using the original images.

The model receives the original leaf and background information.

Strategy B: Background-Randomized Training

Train the model using images in which the original background has been replaced by randomly selected backgrounds.

The disease label remains unchanged because the leaf and its symptoms are preserved.

Strategy C: Mixed Background Training

Train the model using a mixture of:

- Original images.
- Leaf-only images.
- Random-background images.
- Background-modified images.

This approach exposes the model to multiple environmental conditions.

Comparison

The three training strategies will be evaluated using:

1. Original test images.
2. Leaf-only test images.
3. Background-replaced test images.
4. Background-modified test images.
5. Independently collected test images.

The objective is to identify whether background randomization improves robustness without substantially reducing normal classification performance.

Stage 6: Explainable AI Analysis

Explainable AI techniques will be used to investigate which regions of the image influence model predictions.

Grad-CAM

Grad-CAM will be applied to generate heatmaps showing image regions associated with a model's prediction.

The heatmaps will be examined under:

- Original images.
- Leaf-only images.
- Background-replaced images.

The main questions are:

1. Does the model focus on the leaf?
2. Does it focus on visible disease lesions?
3. Does its attention shift when the background changes?
4. Does background randomization change the model's attention patterns?

Integrated Gradients

Integrated Gradients can provide an additional feature-attribution analysis.

It will be used to investigate whether the model's predictions depend on leaf pixels or contextual pixels.

Important Limitation

Grad-CAM or Integrated Gradients alone cannot prove that a model recognizes disease symptoms.

A model may focus on the leaf but still classify diseases using leaf shape, general discoloration, or other non-diagnostic characteristics.

To make stronger claims about disease-symptom recognition, a subset of images should receive expert-annotated disease lesion masks.

These masks can be used to evaluate whether model explanations overlap with actual diseased regions.

Stage 7: External Validation

The final stage will investigate whether the model performs well on independently collected grape leaf images.

New images should be collected using different:

- Backgrounds.
- Mobile phone cameras.
- Lighting conditions.
- Leaf orientations.
- Image distances.
- Environmental settings.

The disease categories should be verified by a qualified plant pathology expert where possible.

These images must not be used during training or model selection.

They will be reserved for external evaluation.

If another public grape disease dataset is used, its disease categories must first be mapped carefully to the existing four categories. Datasets with incompatible disease labels should not be treated as direct four-class test sets.

11. Experimental Design

The experiments will be organized into four major groups.

Experiment 1: Original versus Leaf-Only Classification

Research question: Can the model recognize grape leaf diseases without the original background?

Procedure:

1. Train the model using the existing training dataset.
2. Evaluate it using original test images.
3. Generate leaf-only versions of the same test images.
4. Evaluate the trained model again.
5. Compare classification metrics.

Expected interpretation:

If performance remains relatively stable, the model may be using useful information from the leaf.

If performance decreases significantly, further investigation will be needed to determine whether the removed background, segmentation artifacts, or missing contextual information caused the decrease.

Experiment 2: Background-Only Classification

Research question: Can the background alone predict the disease category?

Procedure:

1. Generate background-only images using segmentation masks.
2. Remove the complete leaf region.
3. Train a separate classifier using background-only training images.
4. Evaluate it using background-only testing images.

The background-only classifier should be trained and evaluated using the same group-aware split structure.

A high balanced accuracy would suggest that the background contains class-associated information.

However, it would not automatically prove that the original classifier uses that information.

Experiment 3: Background Replacement

Research question: Does changing the background alter the prediction of the same grape leaf?

Procedure:

1. Select an original test image.
2. Preserve its leaf region.
3. Replace the background.
4. Generate several versions using different backgrounds.
5. Pass every version through the same trained classifier.
6. Compare predicted classes and confidence scores.

This experiment is particularly important because the disease symptoms remain unchanged.

A robust model should ideally produce consistent predictions when only the background changes.

Experiment 4: Background Randomization

Research question: Can randomized backgrounds reduce background dependence?

Procedure:

1. Train a baseline model with original images.
2. Train a second model using randomized backgrounds.
3. Maintain the same architecture and comparable training configuration.

4. Evaluate both models on the same test conditions.
5. Compare accuracy, macro-F1, balanced accuracy, and prediction consistency.

The experiment will determine whether background randomization improves robustness.

12. Evaluation Metrics

Multiple evaluation metrics will be used because classification accuracy alone is insufficient.

12.1 Accuracy

Accuracy measures the proportion of correctly classified images.

Accuracy = Correct Predictions / Total Predictions

It will be calculated for every image condition.

12.2 Precision

Precision measures how many predictions for a particular disease category are correct.

Precision = TP / (TP + FP)

12.3 Recall

Recall measures how many actual examples of a disease category are correctly identified.

Recall = TP / (TP + FN)

12.4 F1-Score

F1-score combines precision and recall.

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Macro-F1 will be emphasized because the dataset contains class imbalance.

12.5 Balanced Accuracy

Balanced accuracy calculates the average recall across all disease categories.

It is particularly useful when one class contains significantly fewer images than another.

12.6 Background Sensitivity

The difference between a model's performance on original and background-modified images can be used to measure sensitivity to background interventions.

For example:

Background Sensitivity = Accuracy(original) − Accuracy(background-modified)

A larger positive difference indicates a larger performance reduction under the selected background modification.

This metric should be reported separately for each transformation.

12.7 Prediction Consistency

Prediction consistency measures how frequently the same leaf receives the same predicted disease category under different background conditions.

Consistency = Number of unchanged predictions / Number of paired evaluations

For example, if the same leaf is evaluated against five different backgrounds and the model produces the same disease prediction in four cases, its consistency for that sample is 80%.

12.8 Confidence Variation

The predicted class probabilities will also be compared.

Even when the predicted class remains unchanged, confidence may change substantially.

Therefore, confidence differences should be investigated alongside classification accuracy.

12.9 Confidence Intervals and Statistical Testing

The study will report 95% confidence intervals using bootstrap resampling.

The resampling unit should be the independent leaf or image group rather than individual near-duplicate images.

Paired statistical tests, such as McNemar's test, can be used to compare classification errors under paired conditions.

Multiple-comparison correction should be applied where appropriate.

13. Proposed Result Tables

The following tables should be populated after completing the proposed experiments.

They are templates, not existing experimental results.

Table 1: Classification Performance Across Image Conditions

Model	Original	Leaf-Only	Background-Only	Background-Replaced
ResNet-18	TBD	TBD	TBD	TBD
SqueezeNet	TBD	TBD	TBD	TBD
EfficientNet-B0	TBD	TBD	TBD	TBD
ViT-Tiny	TBD	TBD	TBD	TBD
DeiT-Tiny	TBD	TBD	TBD	TBD
AGCTNet	TBD	TBD	TBD	TBD

Report accuracy and macro-F1 separately.

Table 2: Prediction Consistency

Model	White Background	Soil Background	Grass Background	Random Background
ResNet-18	TBD	TBD	TBD	TBD
EfficientNet-B0	TBD	TBD	TBD	TBD
ViT-Tiny	TBD	TBD	TBD	TBD
AGCTNet	TBD	TBD	TBD	TBD

Table 3: Training Strategy Comparison

Training Strategy	Original Test	Leaf-Only Test	Random Background Test
Standard Training	TBD	TBD	TBD
Background-Randomized Training	TBD	TBD	TBD
Mixed Background Training	TBD	TBD	TBD

Table 4: Explainability Evaluation

Model	Leaf Region Attribution	Lesion Region Attribution	Background Attribution
ResNet-18	TBD	TBD	TBD
EfficientNet-B0	TBD	TBD	TBD
ViT-Tiny	TBD	TBD	TBD
AGCTNet	TBD	TBD	TBD

The attribution measurements should use appropriate ground-truth masks and validated explanation metrics.

14. Expected Outcomes

The research is expected to provide evidence about how different image regions influence grape leaf disease classification.

Several possible outcomes are anticipated.

Outcome 1: Strong Leaf-Only Performance

If leaf-only images maintain strong classification performance, this would suggest that the model can extract useful disease-related information from the leaf itself.

However, lesion-specific evidence would still be needed to demonstrate that the model recognizes the actual disease symptoms.

Outcome 2: High Background-Only Performance

If background-only classification performs substantially better than the appropriate chance baseline, this would indicate that the dataset contains background-related class information.

This would motivate further investigation into dataset acquisition bias and contextual shortcuts.

Outcome 3: Prediction Changes After Background Replacement

If predictions change when the background changes while the leaf remains identical, this would provide direct evidence of sensitivity to background interventions.

Outcome 4: Improved Robustness Through Background Randomization

If background-randomized training improves prediction consistency and maintains classification performance, it would provide evidence that background diversification is a useful mitigation strategy.

Outcome 5: Differences Between Architectures

The experiments may reveal that CNNs, Transformers, and hybrid models respond differently to background modifications.

The strongest architecture should be selected based on performance across multiple experimental conditions rather than original test accuracy alone.

15. Potential Research Contributions

The following contributions are targeted rather than already established.

Contribution 1: Background Dependence Analysis for Grape Leaf Disease Classification

A systematic investigation of background-related predictive information in the NGLD dataset.

Contribution 2: Controlled Foreground-Background Evaluation

A paired evaluation framework using original, leaf-only, background-only, and background-replaced images.

Contribution 3: Architecture-Level Comparison

An evaluation of CNN, Vision Transformer, and hybrid architectures under controlled background interventions.

Contribution 4: Background Randomization Strategy

An investigation of whether randomized backgrounds can reduce contextual dependence.

Contribution 5: Disease-Symptom-Oriented Explainability

An evaluation of whether model explanations overlap with expert-annotated disease lesions.

Contribution 6: Robustness Evaluation

An assessment of whether trained models maintain classification performance under different backgrounds and image acquisition conditions.

16. Limitations and Challenges

Several limitations must be considered.

16.1 Dataset Size

The cleaned dataset contains 2,330 images.

This may be insufficient to represent the complete diversity of grape disease symptoms and environmental conditions.

16.2 Class Imbalance

The Bacterial Leaf Spot category contains only 100 original images and 11 images in the current test set.

Its evaluation results will therefore have substantial uncertainty.

More samples are needed.

16.3 Segmentation Errors

Automatic segmentation algorithms may fail to distinguish diseased regions from the surrounding background.

Incorrect masks could remove important disease symptoms.

A manually verified subset is necessary.

16.4 Background Replacement Artifacts

Poorly generated background replacements may create unnatural boundaries, shadows, or color inconsistencies.

A model may respond to these artificial artifacts rather than the intended background change.

Careful compositing and quality checks are necessary.

16.5 Leaf-Level Versus Symptom-Level Recognition

A model may classify diseases using leaf shape, overall discoloration, or other leaf characteristics without recognizing the actual lesions.

Leaf-only classification is therefore not sufficient to establish symptom recognition.

Expert lesion annotations would strengthen the research.

16.6 External Generalization

The NGLD dataset comes from a particular geographic region.

Performance on this dataset does not establish generalization across other grape-growing regions.

Independent data collection is required.

16.7 Statistical Uncertainty

Small test samples, especially for minority classes, can produce unstable performance estimates.

Confidence intervals and group-aware evaluation are necessary.

16.8 Research Novelty

Background bias and domain generalization in plant disease classification have already been studied.

Therefore, the research contribution must be established through its specific grape disease experiments, controlled evaluation, and demonstrated findings.

A new architecture is not necessary to make the study valuable.

17. Ethical and Practical Considerations

The research should follow several principles:

1. Disease labels should be verified against reliable dataset documentation.
2. Newly collected images should be correctly labeled.
3. Dataset sources and licenses should be acknowledged.
4. Test data should not be used for model selection.
5. Results should not be manipulated to achieve higher accuracy.
6. Negative experimental results should be reported.
7. Model explanations should not be treated as proof of medical or agricultural diagnosis.
8. The final system should be presented as a research prototype rather than a replacement for professional plant pathology assessment.

18. Proposed Implementation Tools

Task	Recommended Technology
Programming	Python
Model development	PyTorch
Image processing	OpenCV
Image segmentation	SAM, GrabCut, U-Net
Data analysis	NumPy, Pandas
Visualization	Matplotlib
Explainable AI	Grad-CAM, Captum
Evaluation	Scikit-learn
Statistical analysis	SciPy, Statsmodels
Experiment tracking	MLflow or Weights & Biases
Deployment	Streamlit or FastAPI
Version control	Git and GitHub

19. Conclusion

This research investigates an important limitation of deep learning-based grape leaf disease classification: the possibility that a model may depend on background and contextual information instead of learning sufficiently reliable disease-related visual characteristics.

The existing grape leaf classification experiments have already demonstrated high predictive performance across multiple CNN, Transformer, and hybrid architectures.

However, the preliminary background-bias experiment identified substantial disease-associated information in image regions outside the central area. The corners-only classifier achieved 85.90% balanced accuracy, indicating that contextual information deserves further investigation.

This result does not prove that the original deep learning classifiers depend on background information. Instead, it establishes a motivation for more controlled experiments.

The proposed methodology will evaluate original images, leaf-only images, background-only images, background-replaced images, and background-randomized training conditions.

It will also incorporate image segmentation, explainability, paired prediction analysis, and external validation.

The ultimate goal is to determine whether grape leaf disease classifiers can maintain reliable predictions when environmental information changes and whether background randomization can improve their robustness.

The central objective is to move beyond high classification accuracy toward trustworthy, disease-focused, and environmentally robust AI systems for agricultural applications.

20. References

1. Geirhos, R., Jacobsen, J.-H., Michaelis, C., et al. (2020). *Shortcut learning in deep neural networks*. Nature Machine Intelligence, 2, 665–673. https://doi.org/10.1038/s42256-020-00257-z
2. Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). *Using Deep Learning for Image-Based Plant Disease Detection*. Frontiers in Plant Science, 7, 1419. https://doi.org/10.3389/fpls.2016.01419
3. Beery, S., Van Horn, G., & Perona, P. (2018). *Recognition in Terra Incognita*. Proceedings of the European Conference on Computer Vision (ECCV), 456–473.
4. Dharrao, M., Dharrao, D., Sonawane, R., & Zade, N. (2025). *Niphad Grape Leaf Disease Dataset (NGLD)*. Mendeley Data, Version 5. https://doi.org/10.17632/8nnd2ypcv3.5
5. *Domain generalization plant leaf disease recognition: Toward from laboratory to field*. (2025). Engineering Applications of Artificial Intelligence, 156, 111168. https://doi.org/10.1016/j.engappai.2025.111168
6. Wspanialy, P., & Moussa, M. (2020). *Towards automatic field plant disease recognition*. Computers and Electronics in Agriculture. Publisher article

21. Final Research Direction

The research should ultimately answer three interconnected questions:

1. Detection: Does the dataset contain background-related information that can predict disease labels?

2. Dependence: Do the actual trained grape disease classifiers change their predictions when backgrounds are removed or replaced?

3. Mitigation: Can background randomization and foreground-focused training improve classification robustness without sacrificing disease recognition performance?

Answering these questions with controlled experiments, independent validation, and transparent reporting will provide a stronger research contribution than simply proposing another high-accuracy classification architecture.